SMART-G demo notebook: what it computes and what it gave

Results of *demo_notebook.py* from the SMART-G repository (hygeos/smartg, commit 0c04756), run on this machine (NVIDIA GeForce RTX 3090, CUDA 13.1, pycuda 2026.1, pytrunc 2.1.0). All 69 code cells ran to the end without an error and produced 53 figures; the cells took about 8 minutes of compute in all. Every number and figure below comes from that run, except the few I mark as read from a figure.

1. What the notebook is

This is the general tutorial of SMART-G, a Monte Carlo radiative-transfer code that runs on the GPU. It computes how sunlight is scattered and absorbed in the atmosphere, reflected by the surface and, optionally, scattered in the sea, and it returns the light as Stokes vectors (I, Q, U, V), so polarisation is included. The notebook walks through the features one by one, in the order below. Each section of this document corresponds to a section of the notebook.

How to read the figures. The standard SMART-G view (*smartg_view*) shows a polar map of the field over the sky: the radius is the zenith angle, the angle is the azimuth. Under each map is a cut through it along one azimuth plane; negative zenith angles are the other side of that plane (the dashed part). The second map is the degree of polarisation (DoP, 0 to 100%). The main outputs are reflectances (the radiance normalised by the incoming solar flux), so they have no unit, and the sun is in the direction given by *th_deg* (solar zenith angle).

2. Quick start: Rayleigh atmosphere alone

The first run is the simplest case: a molecular (Rayleigh) atmosphere, the tropical AFGL profile, no surface, wavelength 500 nm, sun zenith 30°, 1e9 photons. It took 28.6 s including the CUDA start-up. The result is the upward reflectance at the top of the atmosphere (TOA) on a grid of 45 zenith × 90 azimuth angles.

Quantity	Value
Reflectance at nadir (zenith 1°, azimuth 20°)	about 0.055 (first values of the row printed in the notebook: 0.0546 0.0601 0.0647 0.0708 0.0779 0.0874 0.1025 0.1270 0.1879 for zenith 1°, 11°, ... 81°)
At zenith 45°, azimuth 90° (interpolated)	0.0608
Largest values	near the horizon (zenith 87°): up to about 0.25
Linearly polarised radiance, largest values	about 0.17, near the horizon at azimuths of 100° and 250°

The sky gets brighter towards the horizon because the light crosses more air. The polarisation is strongest 90° away from the sun, where the DoP reaches about 85% in the cut through the principal plane. The notebook then shows how to pull one variable out of the result with *xarray* (indexing by position or interpolating by coordinate), compute the polarised radiance from Q and U, and plot a 2D map or a transect.

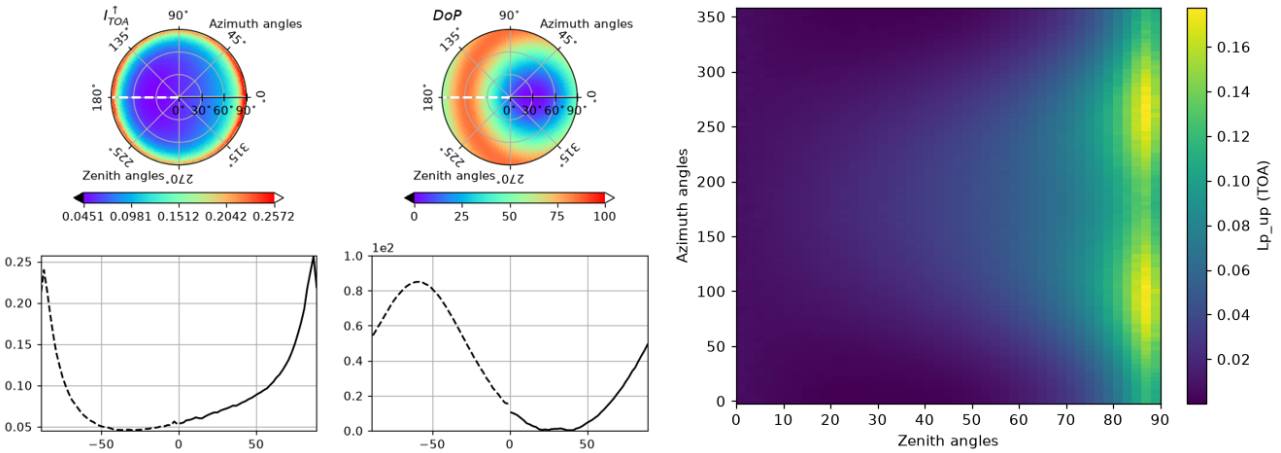


Figure 1. Left: the standard view of the Rayleigh-only run (intensity and degree of polarisation, with cuts). Right: the polarised radiance as a zenith-azimuth map (the notebook's own plot).

A second run with a different random-number generator (CURAND_PHILOX, 34.5 s) and the full view shows all four Stokes components, plus the logarithm of I, the linear and circular polarisation. In the cut through the principal plane U is zero and V is about 10^{-3} or less; U is not zero elsewhere on the map (about ±0.15), as the polarisation direction turns away from the principal plane. Circular polarisation (V) is negligible.

3. Atmosphere, surface and ocean, one layer at a time

The notebook builds the scene up in steps, each a few seconds:

- **Rayleigh with options**: no depolarisation (*depol=0*), a custom vertical grid of 16 levels, surface pressure 990 hPa.
- **Aerosols in the ultraviolet**: urban aerosol (optical depth 0.4 at 550 nm) at 322 nm, sun zenith 60°, looking at the *downward* radiance at the ground (*output_layers=1*). The map shows a strong aureole around the sun's direction (a forward-scattering peak of about 0.95 at the sun's zenith angle).
- **Atmosphere plus a Lambertian surface**, albedo 0.1, 490 nm, sun zenith 45°. Q and U are also drawn.
- **Atmosphere plus rough ocean plus water**: wind 5 m/s, water refractive index 1.34, a 5 m water layer with chlorophyll 0.5 mg/m³ (*HydrosolPR*), at 500 nm and sun zenith 60°. With *output_layers=3* the code gives the light at six places: up at the TOA, up just above the surface (0+), up just below (0−), down above (0+), down below (0−) and down at the bottom of the 5 m layer (B).
- **Surface plus ocean, no atmosphere** and **ocean only**: the same, to see each part alone.

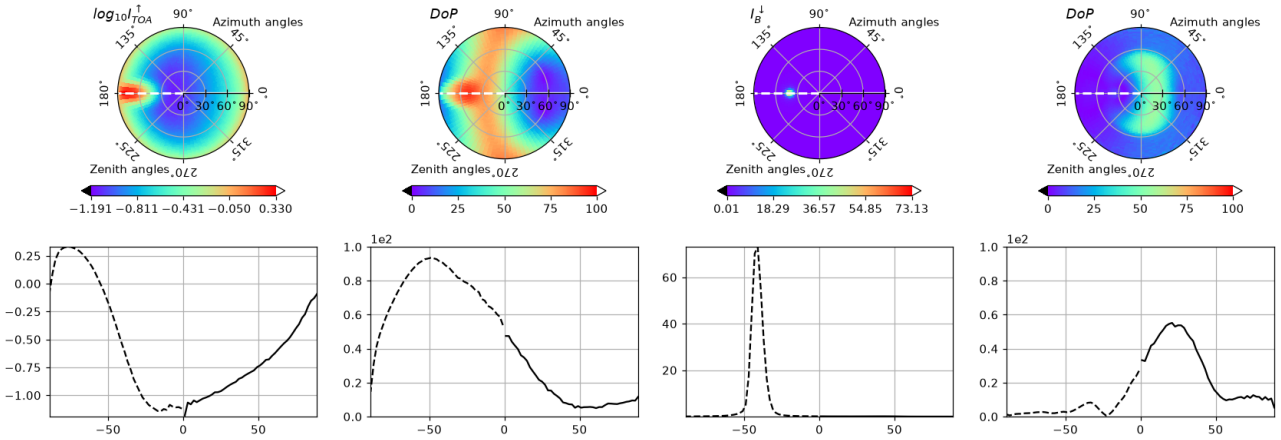


Figure 2. The ocean run. Left: upward radiance at the TOA (log scale); the bright spot at the left is the sun glint on the sea. Right: downward radiance at the bottom of the 5 m water layer: all the light is in a narrow beam, because the refracted sun is the only strong source.

What it shows: at the bottom of the water (right) the light is concentrated in a cone around about 40°, which is where Snell's law puts the refracted sun (sin 60° / 1.34 gives 40.3°; I read the peak off the figure as roughly 40–45°). Above the surface the glint is a small spot near the specular direction. The upward light just below the surface (0−) is ten to a hundred times weaker than the glint (from the colour scales),

which is why the notebook uses logarithmic scales for these fields.

4. Advanced use

4.1 Local Estimate

In the normal mode the result is collected in angular bins, so each bin gets few photons. The *local estimate* (*LocalEstimate*) instead lets every photon contribute to a set of chosen directions, so far fewer photons are needed: this run uses 1e6 photons for 8 zenith × 12 azimuth directions of the upward radiance just below the sea surface (8.6 s). A second form (*zip=True*) takes the angles in pairs, giving a list of 12 directions. The notebook overlays the two results on one polar plot (circles for the grid, squares for the pairs) with the same colour scale, so they can be compared by eye; it is a visual comparison only.

4.2 Custom atmospheric profile

An *Atm1D* object takes the standard AFGL profile and lets you change almost anything: a desert aerosol (optical depth 0.3) mixed with a water cloud (effective radius 11 µm, between 2 and 3 km, optical depth 1), ozone and NO2 switched off, water vapour 2 g/cm², sea-level pressure 980 hPa, a forced Rayleigh optical depth of 0.1, a custom vertical grid with the surface at 1.15 km. *input_view* plots what SMART-G was given: the phase-matrix elements (P11, P12/P11, P33/P11, ...) for each type of particle, and the vertical profiles of the scattering and absorption coefficients. The notebook also shows how to bring your own aerosol phase function from a text file and set the single-scattering albedo (0.80) and the optical depth (0.3262).

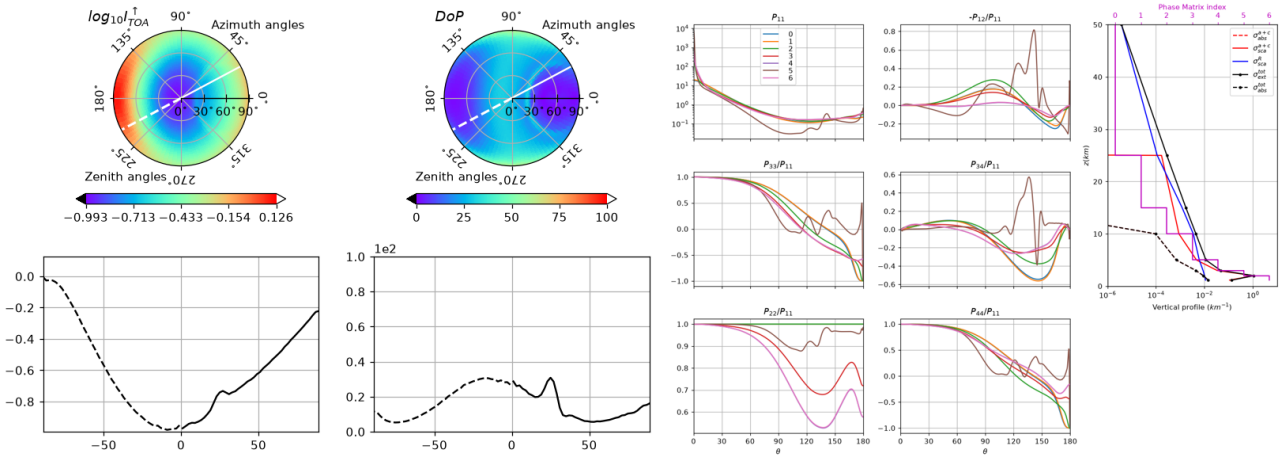


Figure 3. Left: the run with the desert-plus-cloud profile (log intensity and polarisation). Right: the inputs seen with input_view (phase matrices and vertical profiles).

4.3 Truncation of the forward peak (water cloud)

Cloud droplets scatter most of the light almost straight ahead. That forward peak is hard on a Monte Carlo code, because it needs many photons for a small contribution. The notebook reproduces an example from Iwabuchi et al. (2009): a water cloud of optical depth 1, 5 and 20 (droplet radius 8 µm), once with the exact phase function and once with a *GTTrunc* truncation (*trunc_frac* 0.435), at 500 nm, sun at 60°. The plots give the reflected and the transmitted radiance against the viewing angle, on a log scale.

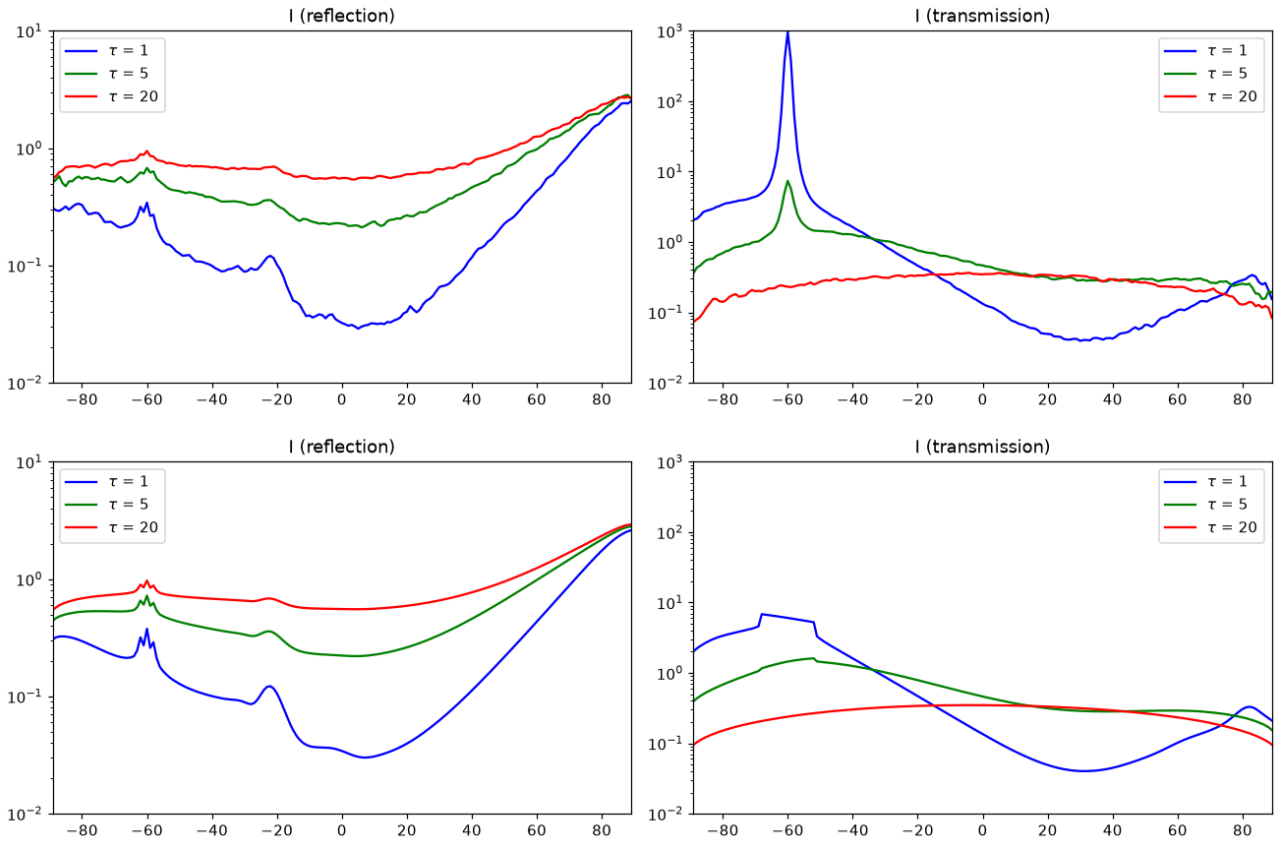


Figure 4. Top: without truncation. Bottom: with the GT truncation. Left panels: reflection; right panels: transmission, for optical depths 1 (blue), 5 (green) and 20 (red).

The two reflection panels look alike. In transmission the sharp peak of the optical-depth-1 cloud, at a viewing angle of −60° (the direct sun, scattered only slightly), is about 1000 in the exact run and about 7 in the truncated one (read from the figures), while the thick clouds (τ = 5, 20) are barely changed. That is the purpose of the truncation: a smooth, cheap-to-sample phase function that keeps the multiply-scattered light correct, at the price of the thin-cloud forward peak.

5. Many wavelengths

5.1 Independent wavelengths, reuse of profiles, and ALIS

The notebook computes the reflectance at 301 wavelengths from 400 to 430 nm, with urban aerosol and NO2 included, at two azimuths. There are three ways, compared in one figure:

Method	Photons	GPU time	What the curve looks like
Independent runs for each wavelength	3e7 in all (1e5 per wavelength)	0.746 s	noisy: the points scatter by about ±0.002 around the trend (read from the figure)
The same, reusing the computed atmospheric profile	3e7	0.818 s	equally noisy (the time is not lower here: the profile is cheap for 301 points)
ALIS (Emde et al., 2011): one set of photon paths shared by all wavelengths, corrected spectrally	1e5 in all	0.496 s	smooth line through the middle of the noisy points, with 300 times fewer photons

ALIS is built for high-resolution spectra, where the absorption changes quickly with the wavelength while the scattering varies slowly. The notebook uses it on the NO2 absorption (*differential optical absorption spectroscopy, DOAS*): the ratio of the spectra with and without NO2 is 0.976–0.986 for the two geometries, and the direct sun-ground-sensor gas transmission, plotted for comparison, is 0.981–0.989, the same wiggles with a smaller depth. The difference between the two is the effect of scattering on the absorption path.

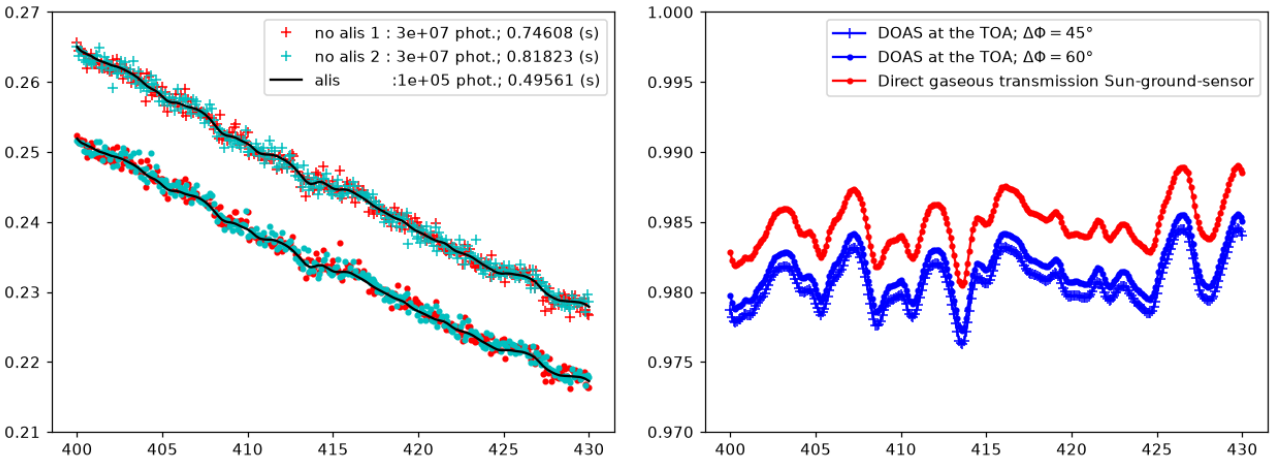


Figure 5. Left: the three methods on the same axes (red/cyan: independent runs; black: ALIS). Right: the NO2 differential absorption at the TOA (blue) against the direct transmission (red).

5.2 A custom water profile

Water1D can take the optical properties layer by layer: the notebook gives the absorption of water, of the particles and of dissolved organic matter, the scattering of water and of the particles, for 450 and 550 nm, over four layers down to 10 m, above a sea floor of albedo 0.05 and with a particle phase function read from a file (1e9 photons, 6.5 s). The upward radiance at the TOA is only about 0.01, and the two wavelengths differ little in shape. *input_view* again plots the phase function and the vertical profiles of the absorption and scattering coefficients.

6. Gas absorption

6.1 One wavelength: two ways of treating absorption

At 550 nm with ozone (300 DU) and NO2, absorption can be counted by lowering the photon weight at each scattering (the single-scattering-albedo method, *beer=0*, the default) or by the Beer-Lambert law along the path (*beer=1*). The two figures are nearly identical: the largest reflectance is 0.3652 in one and 0.3654 in the other (read off the colour scales). They are two ways to the same answer.

6.2 Bands: REPTRAN

For a satellite channel, the absorption by gases varies strongly inside the band, so SMART-G uses *REPTRAN* (Gasteiger et al., 2014): the band is replaced by a handful of representative wavelengths with weights, and *reduce_reptran* sums the results back into a band value. The notebook has four examples.

- **MSG-SEVIRI**: it lists six ways to select bands, then runs the 0.6 µm and 1.6 µm channels (*msg1_seviri_ch006* and *ch016*) over a rough sea at sun zenith 30°. In both channels the sun glint is the main feature (peaks of about 0.23 at 0.6 µm and 0.25 at 1.6 µm); the sky is brighter at 0.6 µm, and nearly black at 1.6 µm.
- **The oxygen A band (757–770 nm)**, polarised reflectance in the sun-glint view, for four sun zenith angles. The polarised signal falls from 0.03–0.14 outside the band (depending on the sun) to 0.005–0.02 at 760–761 nm, where oxygen absorbs; presumably the light that went deep into the atmosphere is absorbed, but I did not test that.
- **Sentinel-3 OLCI**: a break-down for atmospheric correction, with the full scene, only the water, only the surface and no atmosphere (4 runs of 1e8 photons, 54 s).
- **Sentinel-2 MSI**: the 13 spectral response functions are read from a spreadsheet, the REPTRAN coarse solar grid is weighted by them and compared with REPTRAN's own S2 parametrisation.

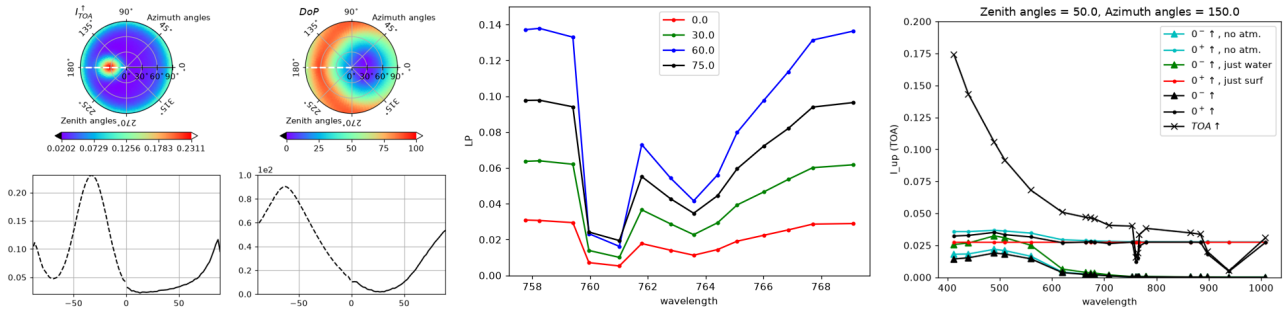


Figure 6. Left: SEVIRI 0.6 µm channel. Middle: polarised radiance across the O2 A band for four sun zenith angles (0° red, 30° green, 60° blue, 75° black). Right: OLCI spectra at the TOA (x) and at the surface (0+, dots) and just under the water (0−, triangles), for the full scene and its parts.

In the OLCI figure the reflectance at the TOA (black crosses) falls from about 0.17 at 412 nm to about 0.03–0.04 beyond 800 nm, which is the blue Rayleigh sky; the sea surface reflection alone (red) is flat at about 0.028; the signal just under the water (triangles, 0−) is 0.015–0.03 in the blue and green and vanishes by 700 nm. Without the atmosphere (cyan) the signal just above the water is the surface reflection plus the water-leaving light. The dips near 760 and 940 nm in the TOA curve are the oxygen and water-vapour absorption bands.

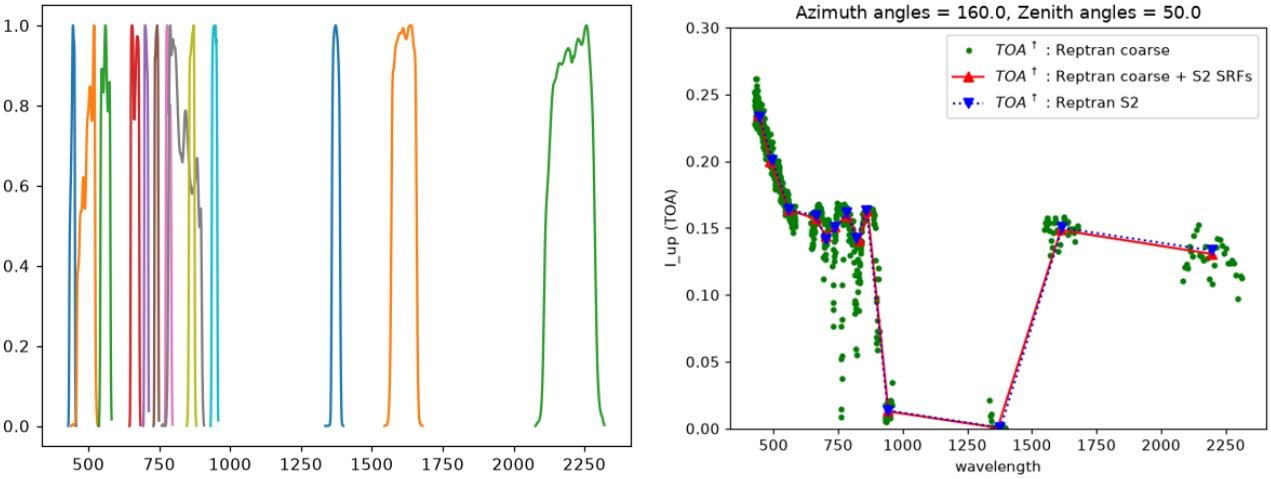


Figure 7. Sentinel-2 MSI. Left: the spectral response functions of the 13 bands. Right: the TOA reflectance at one direction from REPTRAN coarse (green dots), the same weighted by the response functions (red) and REPTRAN's S2 parametrisation (blue): the red and blue lines agree.

7. Surface BRDF

The *RTLSSurface* is a Ross-thick Li-sparse surface, a model of vegetation with an isotropic part and two angular kernels. The notebook uses typical vegetation values: k0 = 0.06 in the visible (440 nm) and 0.36 in the near-infrared (760 nm), with the two kernels at 5% and 30% of k0 (8.3 s with 1e6 photons and a 24 × 24 local-estimate grid). The two figures show how much brighter the scene is in the near infrared: the TOA reflectance reaches 0.39 at 760 nm against 0.27 at 440 nm, and the pattern depends on the view direction as a function of the sun (anisotropic reflection).

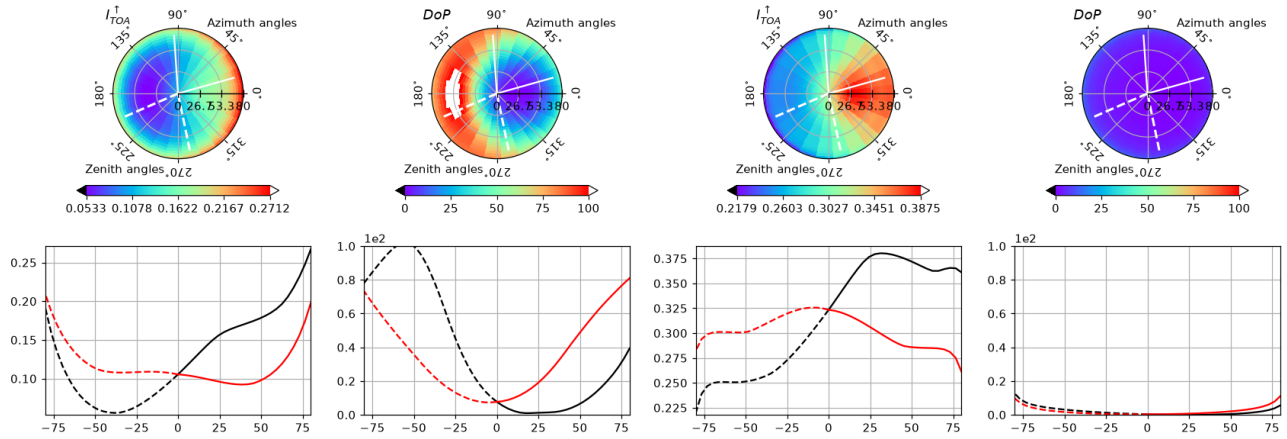


Figure 8. The vegetation BRDF at 440 nm (left) and 760 nm (right), for two azimuth cuts.

8. Irradiances

A flux can be computed in two ways. With *flux*='*planar*' or '*spherical*' SMART-G samples the irradiance directly (fast); otherwise the radiance is computed on a fine angular grid and integrated afterwards with *irradiance_ds*. The test: 11 wavelengths between 400 and 700 nm, sun at 75°, a rough sea with a 100 m water column (chlorophyll 1.1 mg/m³), 1e7 photons for the direct methods and a 360 × 360 angular grid for the integrated one. The irradiance is relative to the incoming solar flux at the top of the atmosphere.

Wavelength (nm)	Planar down (0−)	Spherical down (0−)	Ratio	Planar up (0−)	Direct transmission
400	0.557	0.807	1.45	0.0110	0.245
490	0.675	1.047	1.55	0.0172	0.528
550	0.665	1.046	1.57	0.0150	0.606
640	0.706	1.111	1.57	0.0036	0.732
700	0.780	1.228	1.57	0.0018	0.842

The two ways agree: across all 11 wavelengths the planar flux just under the water from the direct method and from integrating the radiance differ by at most 0.3%, and the spherical flux by at most 0.4%, which is the Monte Carlo noise. The spherical flux (light from all directions on a small sphere) is 1.45 to 1.58 times the planar one (light on a horizontal plane): for a beam refracted to about 46° below the surface 1/cos 46° = 1.44, and the diffuse light pushes the ratio higher, which is consistent with what is seen. The upward flux under the water is small, 0.002 to 0.017, and peaks around 490 nm. The third panel adds the direct beam, which SMART-G keeps apart, to the diffuse downward flux above the surface: diffuse plus direct at 550 nm (planar) is 0.764.

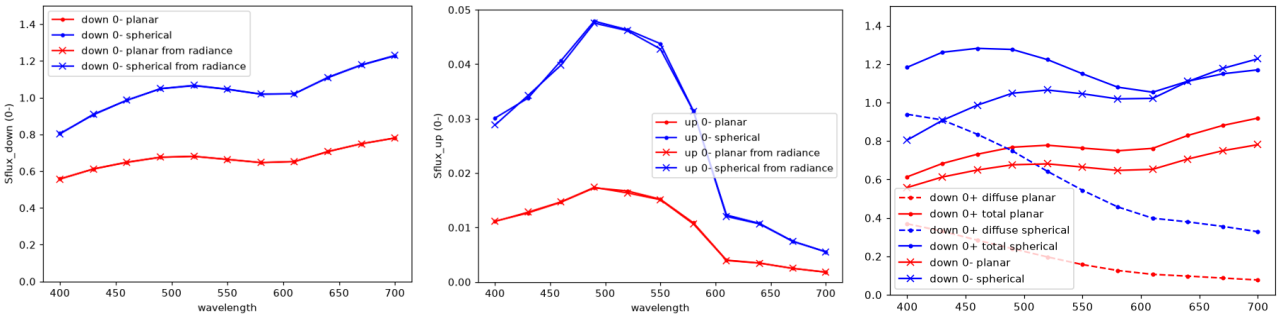


Figure 9. Left: downward flux just under the water. Middle: upward flux. Right: the downward flux above the water, diffuse (dashed) and total (solid). Dots: direct method; crosses: from the radiance.

9. Thermal emission (marked “dev”)

The notebook also runs a thermal-infrared source: the emission of the atmosphere itself, in the thermal channels of the four MSG satellites (platforms 1–4), forward mode only and with no ground, 1e8 photons per platform. The output is the TOA irradiance (W/m²) in 8 thermal channels per platform. Reading the figure: about 0.2–0.5 W/m² at 4–6 µm, 2–4 W/m² between 7 and 9 µm with a dip near 9.7 µm (the ozone band), and 13–15 W/m² in the window channels from 10.6 to 13.4 µm. The four platforms give nearly the same spectrum (within about 1 W/m²). This is marked as a development feature in the notebook, so I treat it as a demonstration.

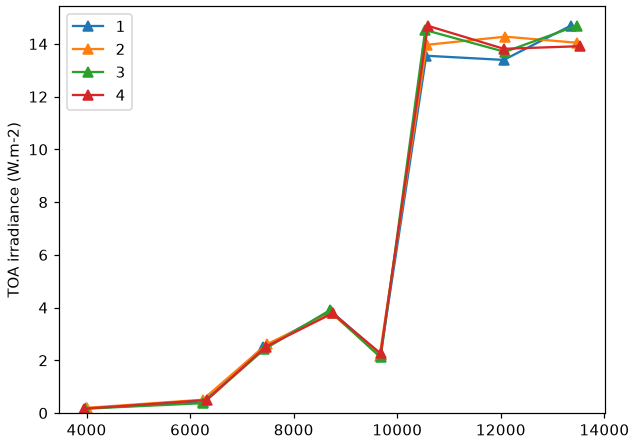


Figure 10. The thermal TOA irradiance of the four MSG satellites' channels (the numbers 1–4).

10. Loops over parameters, and plane-parallel against spherical

Loop over aerosol optical depth. Five runs (1e7 photons each) with a desert aerosol of optical depth 0 to 1.5 at 443 nm, sun at 30°, over a rough sea; the results are stacked along a new *AOT* dimension with *xarray*.

Optical depth at 443 nm	TOA reflectance (zenith 45°, azimuth 90°)	Direct transmission
0.000	0.1069	0.758
0.375	0.1251	0.492
0.750	0.1418	0.319
1.125	0.1560	0.207
1.500	0.1583	0.134

The reflectance rises with the optical depth, then flattens: from 0.107 with no aerosol to 0.125, 0.142, 0.156 and 0.158. The direct transmission falls roughly exponentially, from 0.76 to 0.13. The last two points differ by about 1.5%, similar to the noise of this 1e7-photon run, so the flattening at the end is not established by this run alone. (Figure 11, from the first complete run of the notebook, shows 0.1055 and 0.161 at the two ends; the table comes from a second run made to get exact numbers: a 1–2% difference between runs.)

Plane-parallel against spherical atmosphere (400 nm, sun at 30°, 5e7 photons): the relative difference of the TOA radiance is about +0.3% for viewing zenith angles up to about 65°, and then falls to −1% at 78° and −3% beyond 82° (read from the figure). So the curvature of the Earth matters only for grazing views.

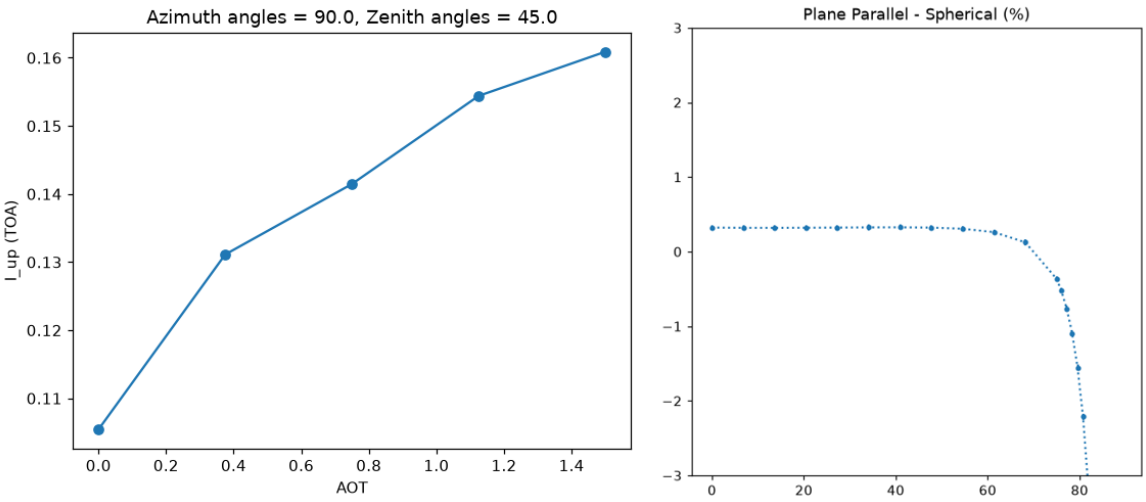


Figure 11. Left: the reflectance against aerosol optical depth. Right: the difference, in %, between a plane-parallel and a spherical atmosphere against viewing zenith angle.

11. The interactive cell

The notebook defines *interact_manual*, a small set of sliders (viewing zenith, surface on or off, aerosol model, optical depth) with a Run button, which simulates one case of 1e5 photons and prints the TOA intensity. Run as a plain script, as I did, the cell only creates the widget: no simulation runs and no value is printed. It needs Jupyter to be used, and I have not tested it there.

12. Backward mode and the Sensor class

In the forward mode used so far, photons start at the sun and the results are collected at the top of the atmosphere. In **backward mode** (*back=True*), photons start at a *Sensor*, which you place anywhere, and the output is for solar directions. It is needed when the scene is not the same everywhere (a spherical atmosphere, a coastline) and for sensors that sit inside the atmosphere. Three examples follow.

12.1 Limb geometry

A sensor at the top of a spherical atmosphere looks at the limb, at 16 tangent heights between 2 and 50 km, 4 relative azimuths, 3 wavelengths (430, 660, 840 nm) and 4 sun zenith angles (60°, 80°, 91°, 94°, with refraction included). The plots show the radiance against tangent height. It falls by about one to three orders of magnitude between 2 and 50 km (less for blue), the blue (430 nm) is the brightest at every height and the red (840 nm) the weakest because molecular scattering weakens with wavelength, and with the sun below the horizon (91° and 94°) the lower curves bend away from the others: the lower atmosphere is shadowed while the high atmosphere is still lit (reading from the figure; no numbers were printed).

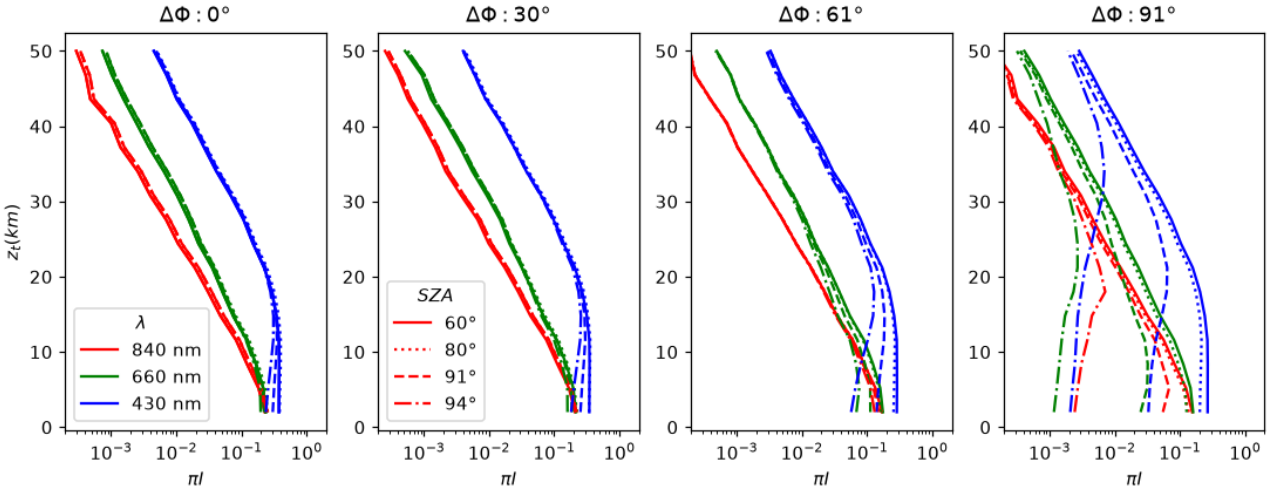


Figure 12. Limb radiance against tangent height, for four relative azimuths (panels), three wavelengths (colours) and four solar zenith angles (line styles).

12.2 A coastline

The scene: sensor at 120 km, viewing zenith 30°, aerosol (maritime polluted, optical depth 0.3), Cox-Munk sea with wind 5 m/s, and a coast between ocean (at x < 0) and snow (albedo 0.5). The coast is modelled as the edge of a huge ocean disk of radius 1e6 km, so it is locally a straight line. The sensor looks at an ocean pixel 250 m, 5 km, or (practically) infinitely far from the coast, from four directions; the plots show the principal-plane reflectance at 380, 500 and 800 nm.

The result: the nearest case (250 m, dotted) is brighter than the homogeneous ocean (solid black), by about 0.05 at 380 nm near the zenith (0.25 against 0.20) and about 0.02 at 800 nm (read from the figures); it is brightest when the sensor looks over the land side. That is the *adjacency effect*: light scattered from the bright snow into the sensor's line of sight. At 5 km (dashed) the curves are close to the homogeneous ocean, so the coast stops mattering after a few kilometres. The polarisation (DoP) differs less than the intensity. (12 sensors, 4 azimuths × 3 distances, 1.2e7 photons, 6.7 s.)

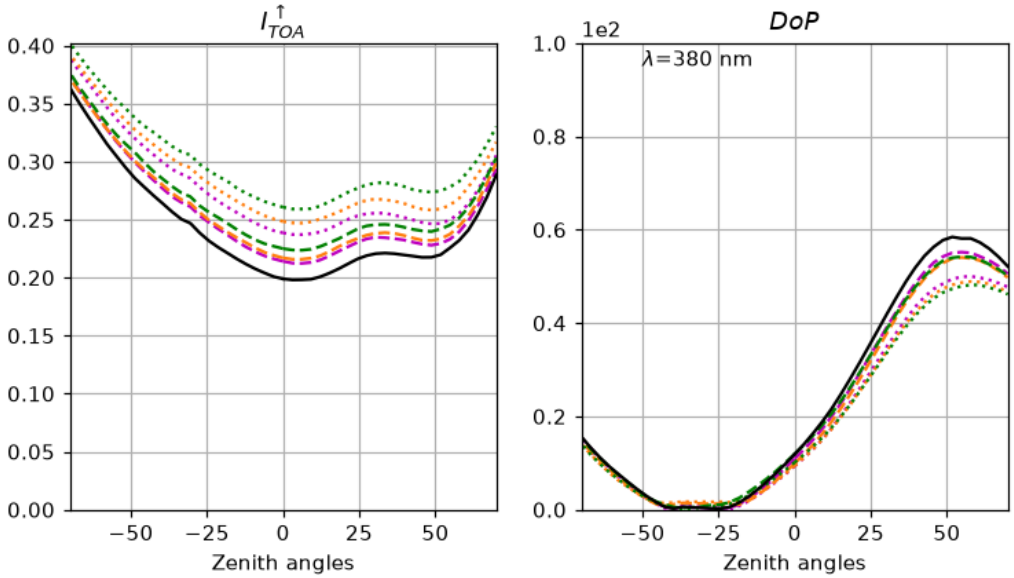


Figure 13. The 380 nm principal plane transect: intensity (left) and degree of polarisation (right). Black: homogeneous ocean; dotted: 250 m from the coast; dashed: 5 km; the colours are the view azimuth relative to the coast (magenta above water, orange along the coast, green above land).

12.3 A 2D albedo map

AlbedoMap defines a rectangular map of different albedos. Here there is a 20 km land square with snow in its western half and soil in its eastern half, surrounded by sea with a 10 m water column over a soil seabed. An 8 × 8 grid of nadir sensors, 120 km up, makes an image at nine wavelengths from 400 to 900 nm (1e8 photons, 7.6 s).

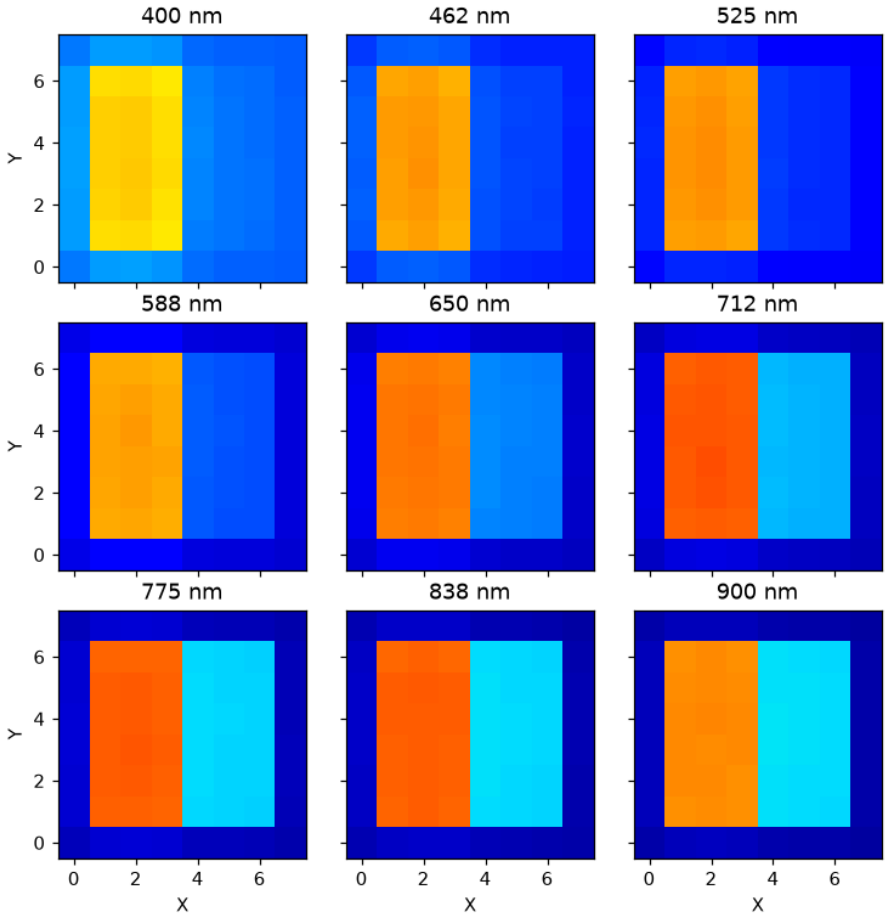


Figure 14. The simulated image at nine wavelengths. The bright left block is snow, the darker right block is soil, and the dark frame is sea. The snow is brightest in the blue and is darker in the near infrared; the soil brightens with wavelength.

The snow is the brightest block at every wavelength; its contrast with the soil is largest in the blue and smallest in the near infrared, where the soil is much brighter than in the blue. The sea frame is the darkest at all wavelengths and almost black in the infrared, where water absorbs everything. Each pixel is 2.9 km wide (8 sensors over 20.2 km). The colour scale is fixed at 0 to 1, so values can only be read off colours.

13. Counters: where the light went

The last section uses the counters *SMIN/SMAX* (minimum/maximum number of interactions), *RMIN/RMAX* (reflections on the surface) and *DIRECT* (count the direct beam), to split the atmospheric transmission (590 nm, US-standard atmosphere, sun at 30°, ozone 300 DU, 1e8 photons per run).

Quantity	Value	Meaning
T	0.91776	total transmission, direct beam included
Tdif (black surface)	0.03823	diffuse transmission only
Tdif_0 (no surface reflection)	0.03818	diffuse transmission counting only photons that were not reflected at the ground (surface albedo 1, *r_max=0*)
Tdif_0_MS	0.00436	of which multiple scattering (*s_min=2*)
Tdif_0_SS	0.03385	of which single scattering (*s_max=1*)

The numbers are consistent: T − Tdif = 0.8795 is the direct beam (a direct transmission of 0.88 at 590 nm, sun at 30°); Tdif_0_SS + Tdif_0_MS = 0.03821, which equals Tdif_0 (0.03818) to 3×10^{-5}; and Tdif_0 equals Tdif of the black surface (0.03823) to 5×10^{-5}, as it must when the surface reflections are excluded. Single scattering is 89% of the diffuse transmission, multiple scattering 11%.

14. Summary and caveats

- Every cell of the notebook ran. The one cell whose point is interactive (the sliders) ran only as far as creating the widget.
- The results are consistent with each other where the notebook allows a cross-check: two ways of counting absorption (section 6.1), two ways of computing fluxes (section 8), the single plus multiple scattering sum (section 13), REPTRAN against response-weighted spectra (6.2), ALIS against the noisy mean (5.1). Where I say a value was read from a figure it is approximate (a few percent).
- **Not checked:** I did not compare against an outside reference, and I did not check the physical values of the OLCI, SEVIRI or thermal figures beyond reading them. The adjacency, limb and thermal comments are my reading of the plots; no numbers for them were printed by the notebook.
- Run-to-run noise: the runs use a new random seed each time. For example the aerosol-loop values differed by 1–2% between two runs. Percent-level differences in the tables are not significant.
- Practical: on this machine the script needs the Visual Studio C++ compiler on the PATH (nvcc), a pytrunc of version 2.1 or later, and the *SMARTG_DIR_AUXDATA* variable pointing to this project's data folder; all three are now in place.
- In the headings above, the section numbers are mine; they differ from the notebook's own headings in a few places because I grouped them.